Local changes in demersal fish biomass in relation to oxygen, temperature, and the metabolic index in a warming and deoxygenating ecosystem

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Running head: Local trends not related to climate

Abstract

The rapid warming and deoxygenation of our oceans is affecting the spatial distribution of organisms. To predict responses across taxa we need general approaches based on individual-level physiology, and to attribute range shifts as responses to climate warming, we need to embrace spatially explicit local scale trends. We analyses more than 4 decades of spatiotemporal trends in biomass of key groundfish species in the southern Baltic Sea — the largest anthropogenically induced hypoxia area and one of the most rapid warming seas in the world — to assess changes species niches, drivers of biomass change, and the support for the metabolic index as a predictor integrating temperature and oxygen effects in a mechanistic framework. We use geostatistical generalized linear mixed effects models estimate thermal niches of 9 species-life stage groups in the southern Baltic Sea, and how these have changed over time and in relation climatic change. Next we estimate spatially explicit spatial trends in biomass from climate-agnostic models, and use machine learning algorithms to estimate the relative importance of variables for explaining these biotic trends, such as the velocity of temperature and mean temperature. In general we find that species thermal niches generally increased at similar rates as the southern Baltic region. Local trends in biomass density were positively associated with temperature, negatively with biomass density, and the spatial patterns varied by species. Lastly, biomass-temperature response curves indicate that on average, these populations experience temperatures below those that maximize biomass in most species. More generally, our results highlight the importance of analyzing local biomass trends and their drivers when attempting to identify mechanisms underlying species' range shifts.

Keywords: climate change, temperature, dissolved oxygen, metabolic index, groundfish, spatio-temporal distribution models, spatial trends

Introduction

1) genera climate change and species re-distribution 2) how are we studying species responses? (recall here we look at local trends, not the common aggregated metric like centre of grvity) 3) how do we deal with oxygen and temperature? 4) baltic case a great case study becuase loosing oxygen and warming, while species cannto mot northwards 5) in this study… we

Methods

Data

Trawl data

Environmental data

Spatiotemporal models

Model fitting

Random forest models

Results

The metabolic index was not a more parsimonious predictor than oxygen concentration and temperature modelled separately, based on conditional AIC. Instead, the most parsimonious model across all species-life stage combinations contains a linear and squared depth effect and a break-point effect of oxygen concentration (Table 1).

Over the last three decades, oxygen concentration declined in the domain by approximately 0.5 ml/L per decade (Fig 2a, c), and temperature increased by 0.5 °C per decade (Fig 2b, d). Trends in the thermal niche of the 7 species-life stage groups, calculated as the linear slope of biomass-weighted temperature, follow the trend in the overall environmental, in that all confidence intervals overlap. This implies any spatial redistribution has not led to thermal niche conservation. Trends in experience oxygen however are more varied, with adult and juvenile cod, and juvenile place, overlapping with the environmental trend, while the remaining species-life stage groups have trends in experienced oxygen somewhere between the trend in the environment and 0 (juvenile and adult flounder in quarter 4), or have confidence intervals overlapping 0.

Conditional effects of temperature coupled with biomass-weighted median temperature and oxygen reveals that adult and juvenile cod, on average, experience temperatures where cod biomass is already maximized, especially in quarter 4 (Fig. 3a–b). All other groups on average occupy temperatures below “optimum” (Fig. 3c–g), and for dab, the conditional-temperature effect is linear within the range of observed temperatures (Fig. S4). Cod, plaice and dab occupy oxygen conditions close to the breakpoint, i.e., the oxygen concentration above which biomass is not predicted to increase with increased oxygen (Fig. 3a, b, e–g; Fig. S5), while flounder occupy habitats relatively high in oxygen on average (approximately 2°C above the estimated breakpoint concentration) (Fig. 3c–d).

When we relate local trends in biomass density (Fig. **??**) to explanatory variables (species group, mean environmental conditions including fish biomass density, environmental velocities) (Fig. **??**) in a random forest model, we find mainly that trends vary across species groups (Fig. **??**). Mean biomass density, mean temperature and mean oxygen are moderately important, and the velocity of temperature and oxygen concentrations are not very important (Fig. **??**). Marginal predictions (partial dependence plots) show that mean temperature is the predictor variable that shows the most coherent pattern across all species-life stage groups, with an overall sigmoidal shape (Fig. **??**; Fig. S3). The effect of mean oxygen is positive for adult plaice and dab (i.e., grid cells with higher oxygen concentrations are associated with stronger trends in biomass density over time), while it is negative for the remaining species (Fig. **??**.

Mean biomass density was only a moderately important variable with, the marginal predictions varied by species (Fig. S3), probability of occurrence showed a cleared relationship between biomass indices and biomass density. Across all species-life stage groups, there was a positive relationship between “fraction area occupied”, defined as the fraction of all grid cells with a probability of encounter > 0.9, and annual indices of biomass (Fig. **??**a) (although the effect if relatively weak in adult cod). In adult flounder and plaice (adult and juvenile), years with low overall biomass and

Discussion

Acknowledgments

We are grateful to everyone involved in data collection. Reviewers.

Author contributions

Data Availability Statement

Code and data to reproduce the results are available on GitHub (https://github.com/maxlindmark/spatial-metabolic-index) and will be deposited on Zenodo.

Funding

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References

Tables

Table 1: Model comparison across fish groups

Group	**temp+oxy**	**temp+temp^2+oxy**	**temp$\times$oxy**	**temp+temp^2+bp(oxy)**	ϕ	**bp(ϕ)**
Cod (A)	646.3	157.7	175.5	**0**	863.1	2950.7
Cod (J)	459.1	93.6	56.3	**0**	804.4	2312.7
Dab	64.2	47.7	29.4	**0**	148.5	553.8
Flounder (A)	907	275.4	64.8	**0**	1853	5001.8
Flounder (J)	207.2	98.4	95.8	**0**	504.5	3565.9
Plaice (A)	141.9	40.4	53.1	**0**	322.3	1490.1
Plaice (J)	120.7	38.4	11.8	**0**	295.9	1424.1

Figures

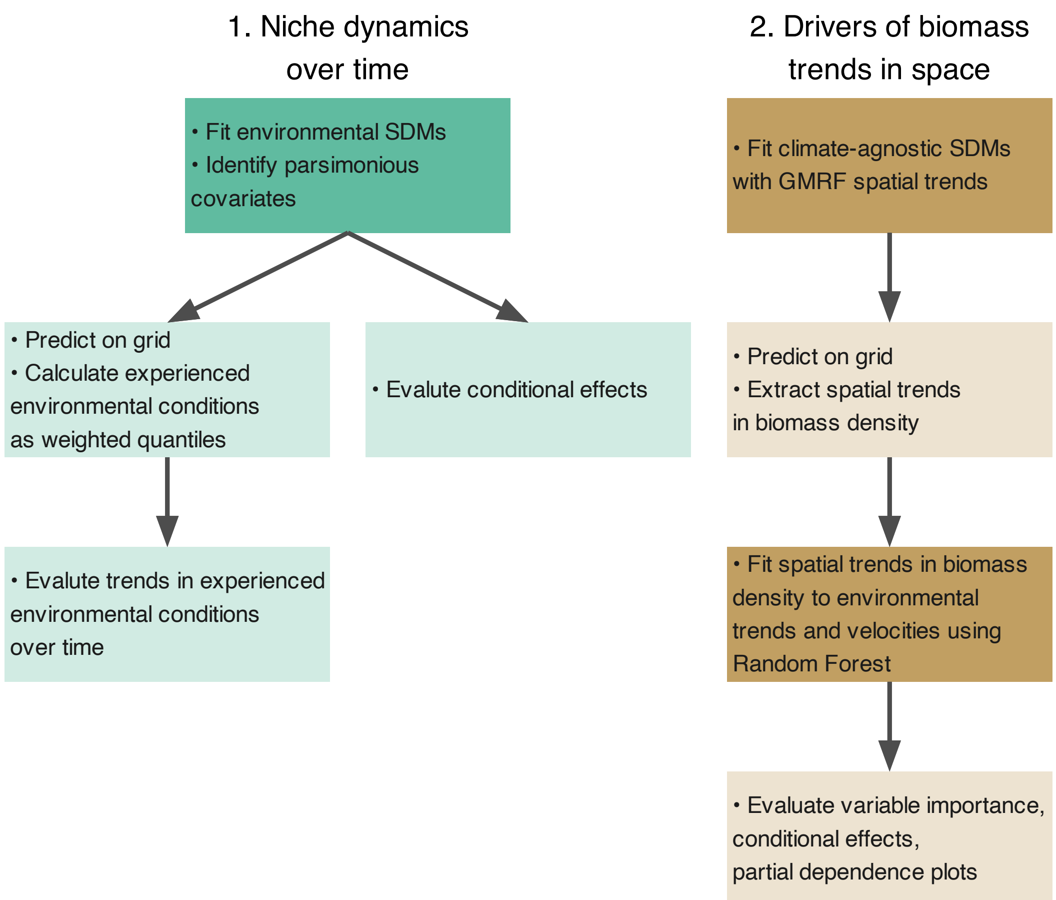

Figure 1: Schematic overview of the modelling steps. The analysis proceeded in three main stages: (1) species distribution modeling to identify parsimonious environmental covariates and evaluate their conditional effects on biomass density (teal boxes), (2) spatial trend modeling using Gaussian Markov Random Fields to extract temporal trends in both experienced environmental conditions and biomass density (orange boxes), and (3) random forest modeling to relate environmental trends to biomass trends and evaluate variable importance through partial dependence analysis. Arrows indicate the sequential flow of data and results between analytical steps.

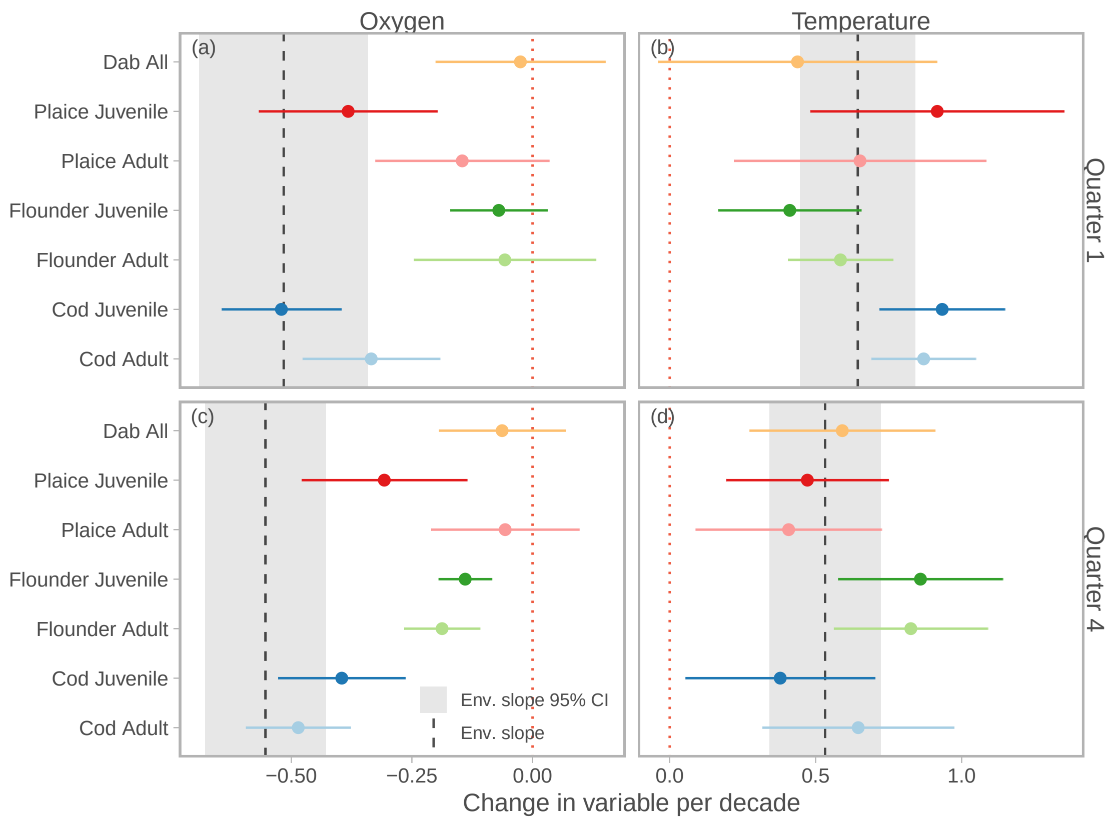

Figure 2: Trends in environments experienced by the species relative to the changes in the environment. Points show slope of year for biomass-weighted oxygen (left column) and temperature (right column) and horizontal lines depict the the 95% confidence interval, for quarters 1 (top row) and 4 (bottom row), for all species-age groups (color). The dashed vertical grey line shows the slope of year for variables in the environment, and the grey rectangle encompasses the 95% confidence interval. The red dashed line is at $x = 0$, which if crossed suggests the trend is not statistically clear.

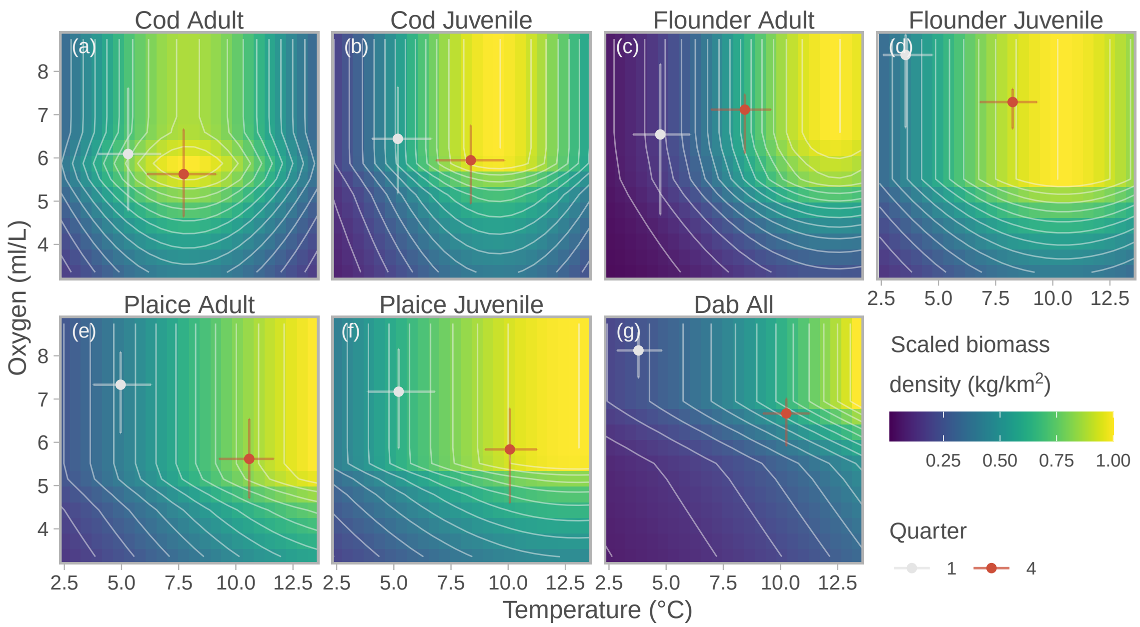

Figure 3: Conditional effects of oxygen (y-axis) and temperature (x-axis) on biomass density (fill color) for species groups (panels). Fill color corresponds to the predicted biomass density, scaled to the maximum biomass density by species to allow comparison across species. White and red points correspond to the biomass-weighted mean temperature and oxygen, for quarter one and four, respectively. Vertical and horizontal lines correspond to the interquartile range of biomass-weighted mean values of oxygen and temperature.

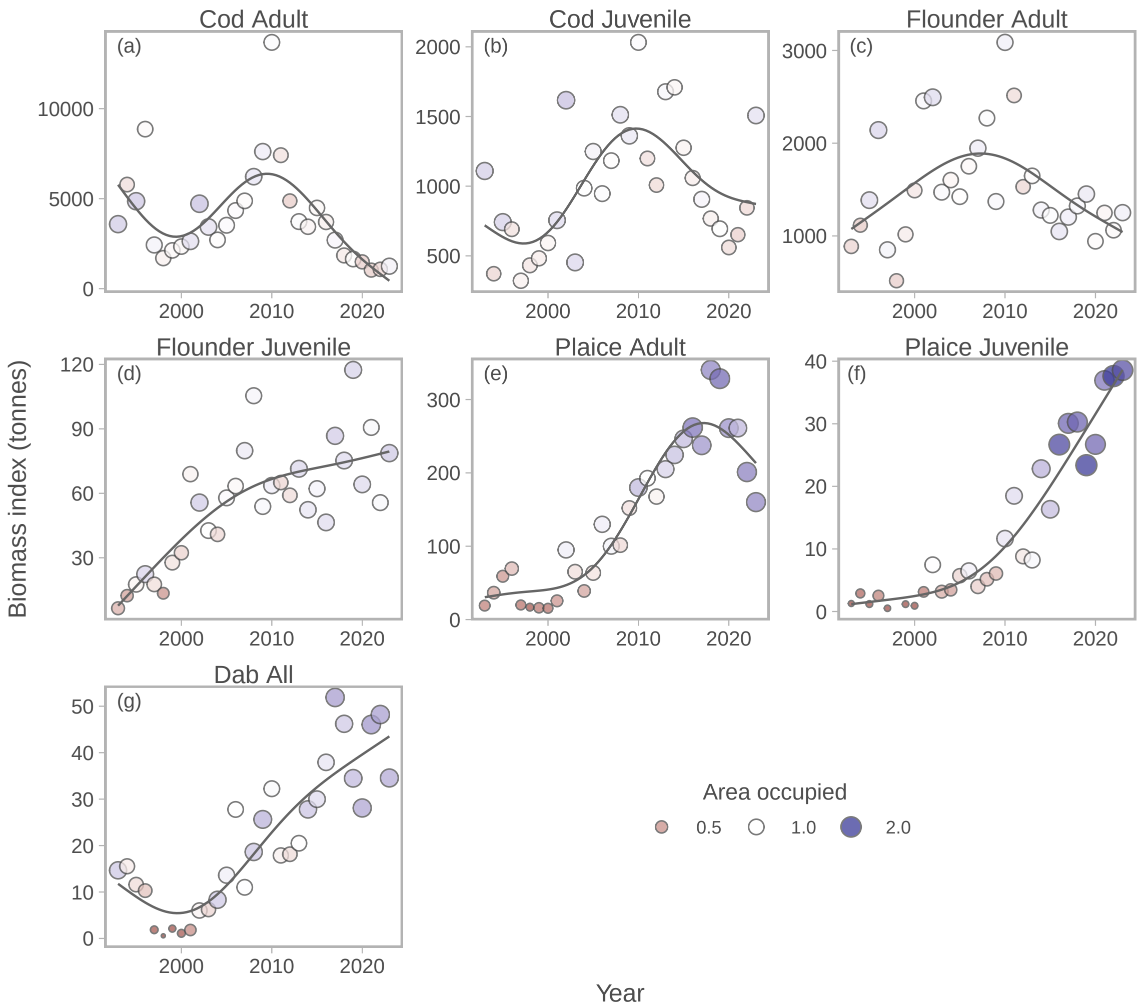

Figure 4: Relationship between biomass indices, time and area occupied, for each species-life stage combination (facet). The y-axis corresponds to the biomass index, x is year, and the fill color and point size corresponds to the relative area occupied (defined as the fraction of all grid cells with a probability of encounter > 0.9), divided by the mean area occupied across all years, by species, to facilitate comparison between species). The thin grey line show the temporal trends, as fits from a generalized additive model with maximum degrees of freedom $k = 5$ for the smooth year effect.

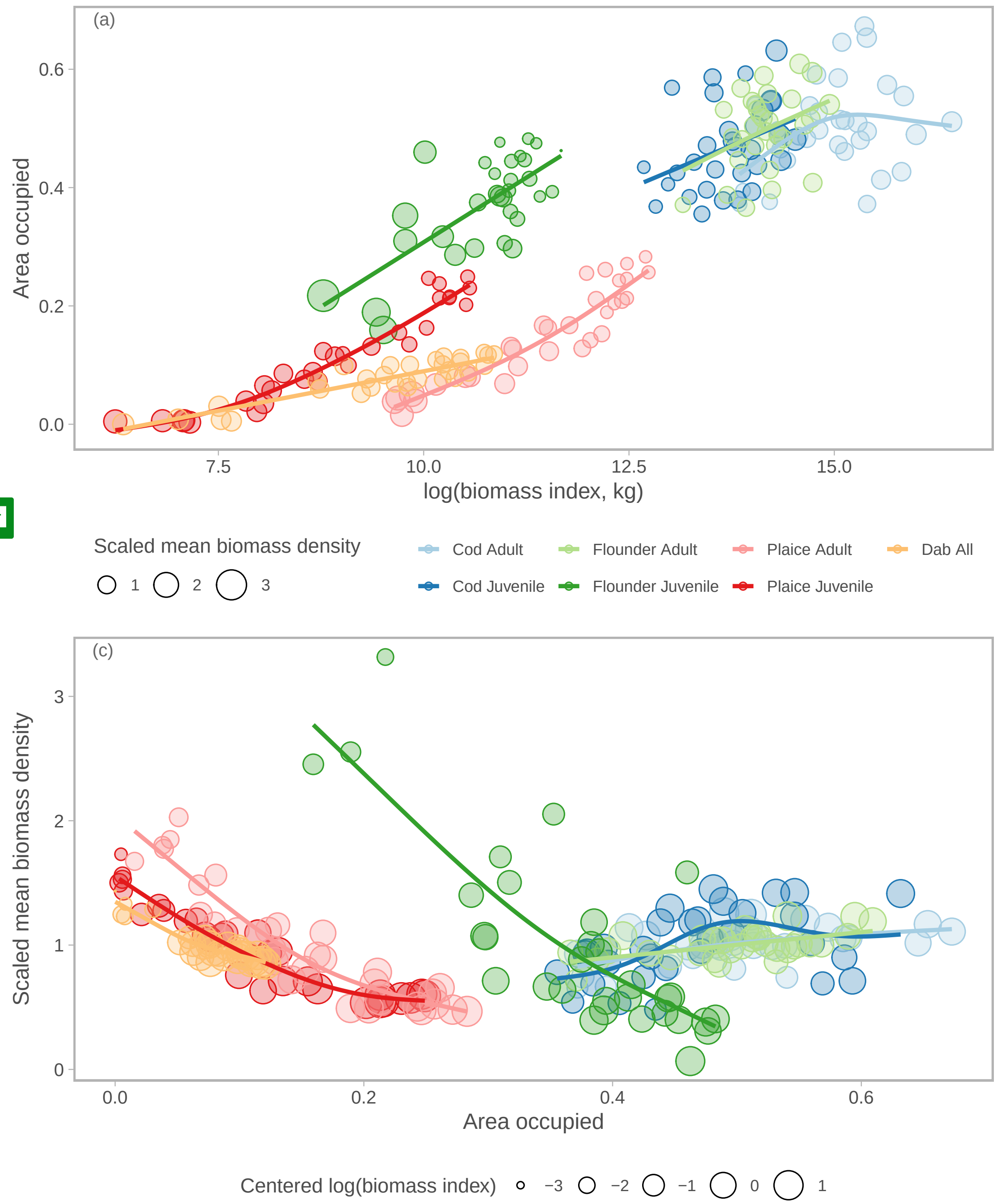

Figure 5: Relationship between area occupied, time, mean biomass density and total biomass (model-based biomass index). Panel (a) shows the relationship between area occupied (defined as the fraction of all grid cells with a probability of encounter > 0.9) and

Supporting Information

S1. Metabolic index

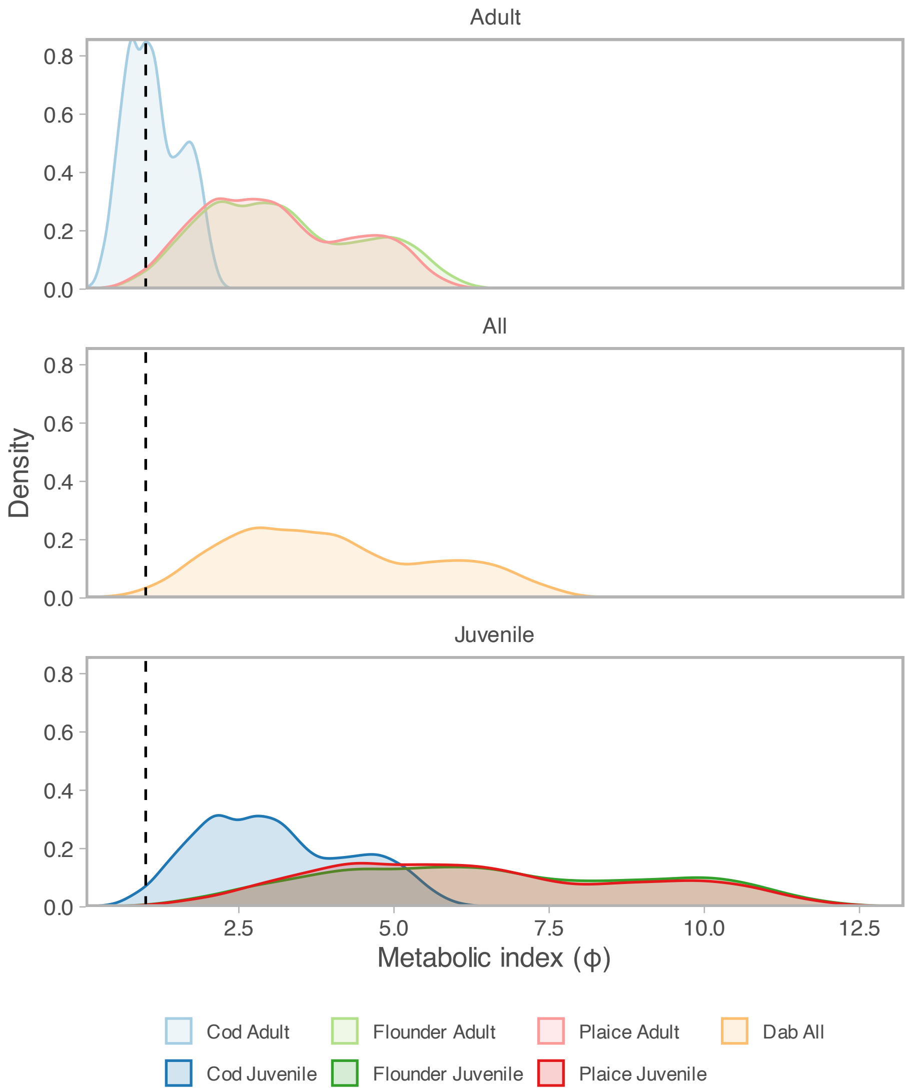

Figure S1: Distribution of metabolic index values.

S2. Spatiotemporal models

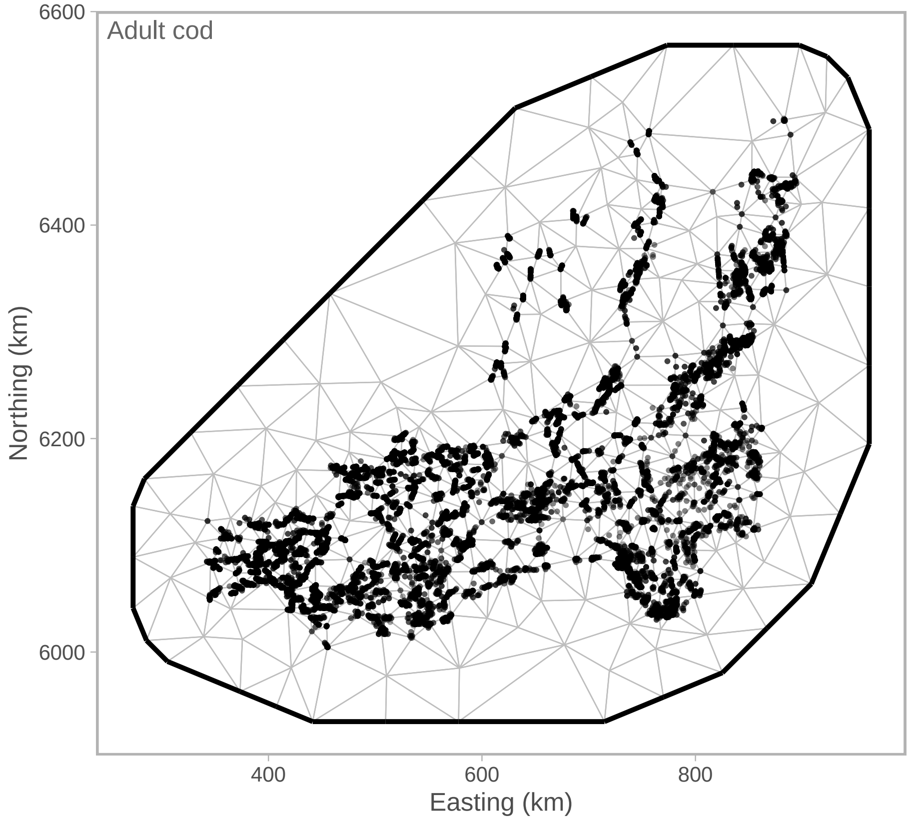

Figure S1: Example SPDE mesh (adult cod).

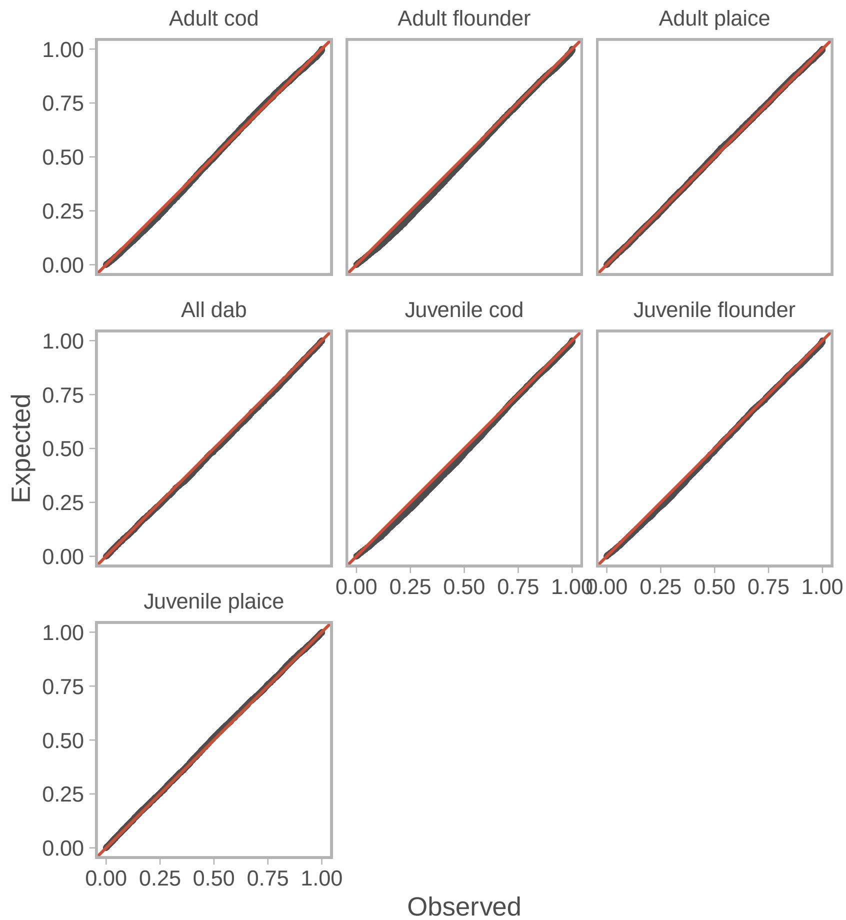

Figure S2: QQ-plots of the Poisson-link delta-gamma biomass density models for comparing AIC and making predictions of conditional effects of temperature and oxygen, based on simulated randomized quantile residuals (Dunn *et al.* 1996, Hartig 2022).

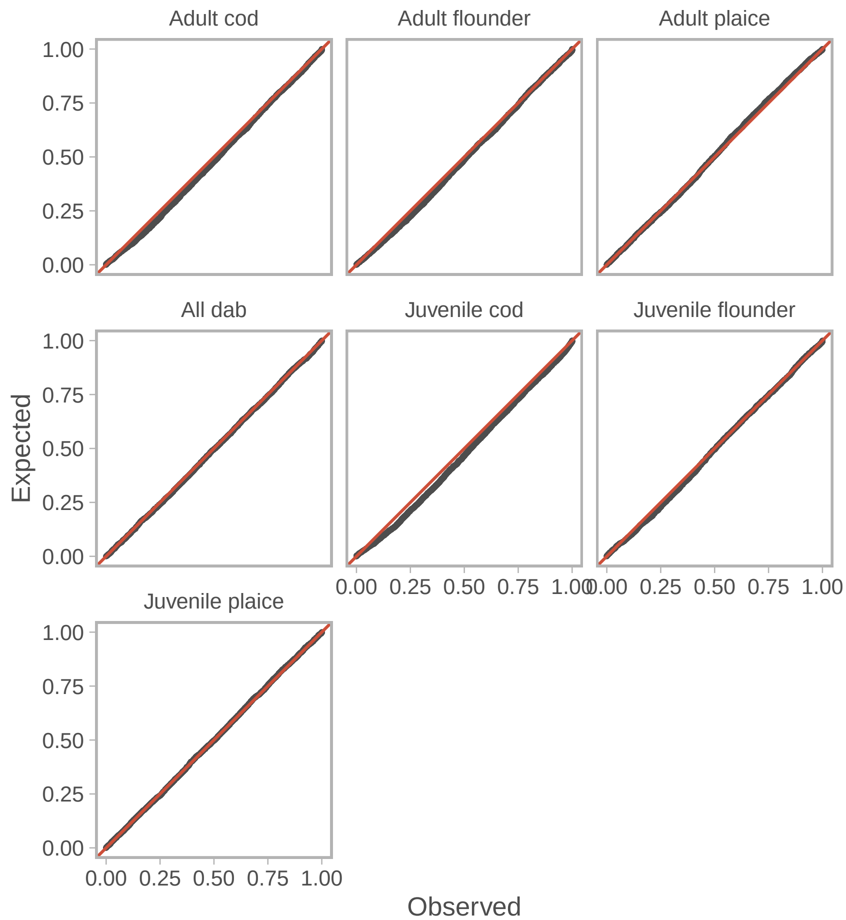

Figure S3: QQ-plots of the climate-agnostic Poisson-link delta-gamma biomass density models for extracting spatial trends in biomass over time, based on simulated randomized quantile residuals (Dunn *et al.* 1996, Hartig 2022).

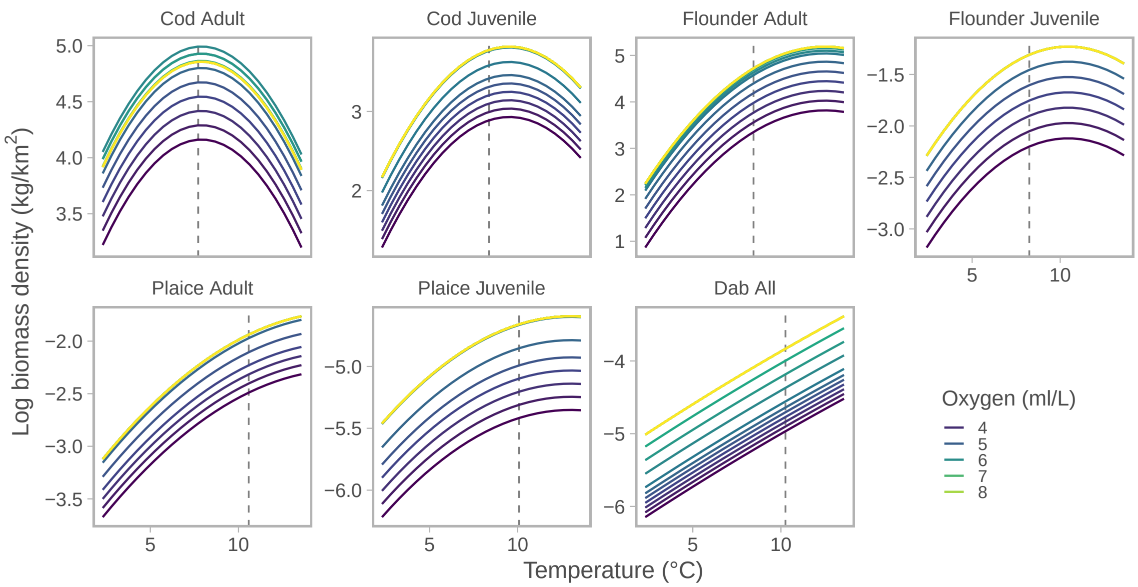

Figure S4: Conditional effects of the quadratic temperature effect in link space, for different oxygen levels (color), holding all other covariates at their mean and omitting random effects.

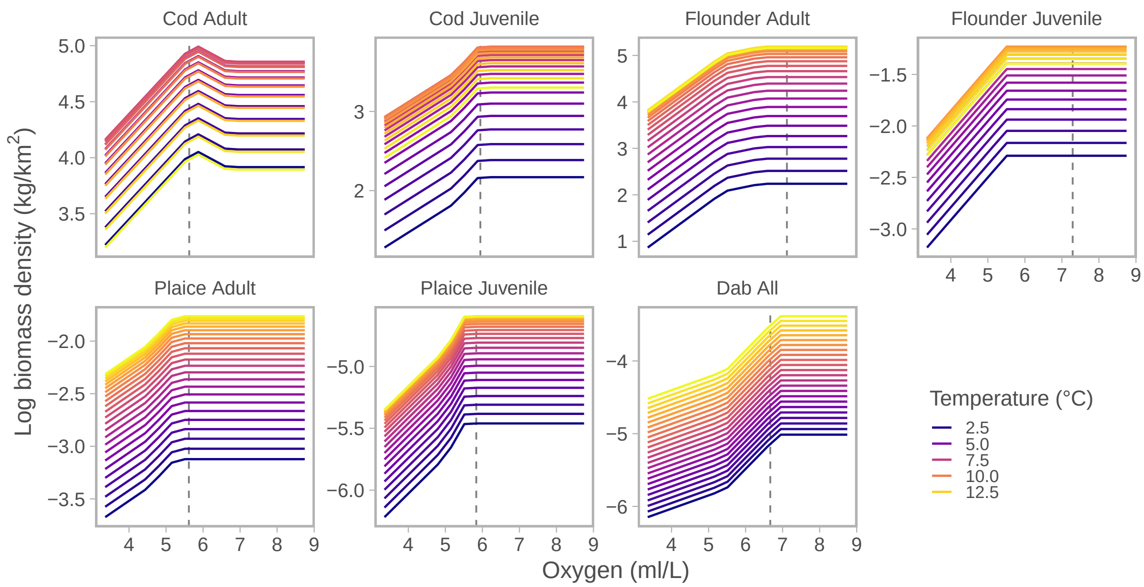

Figure S5: Conditional effects of the break-point (hockey-stick) oxygen effect in link space, for different temperatures (color), holding all other covariates at their mean and omitting random effects.

S3. Random forest model

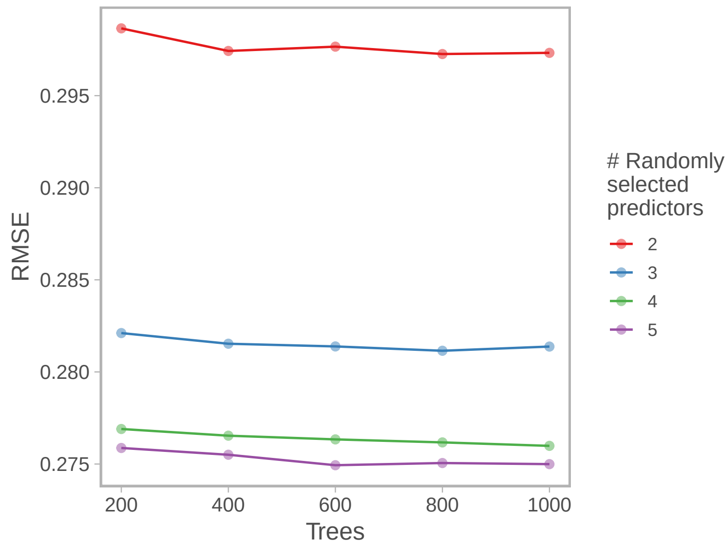

Figure S1: Random Forest hyperparameter tuning results. Cross-validated model performance (RMSE and R^2) for random forest models with varying numbers of trees (200–1000) and randomly selected predictors per split (mtry = 2–5). The configuration minimizing RMSE was selected for final models.

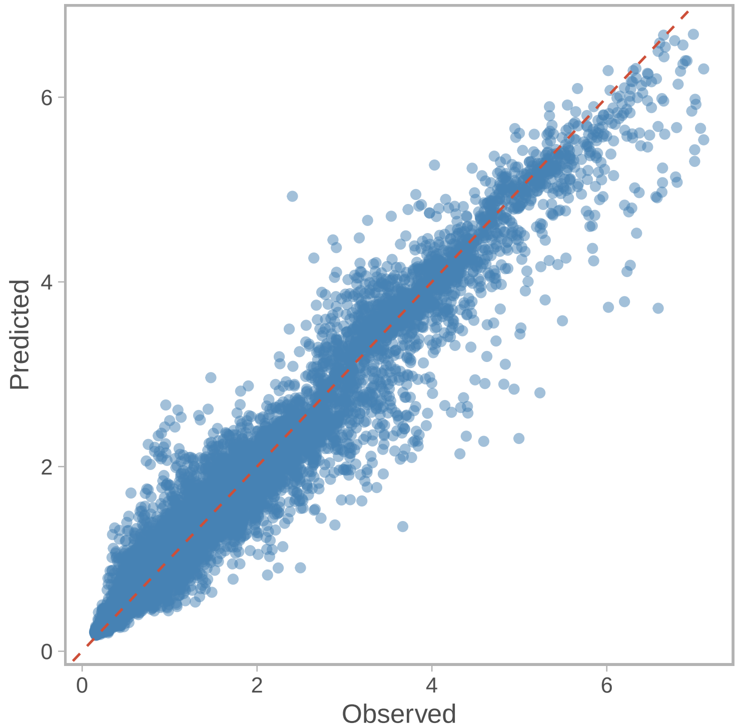

Figure S2: Random forest model predictions versus observed biomass trends.

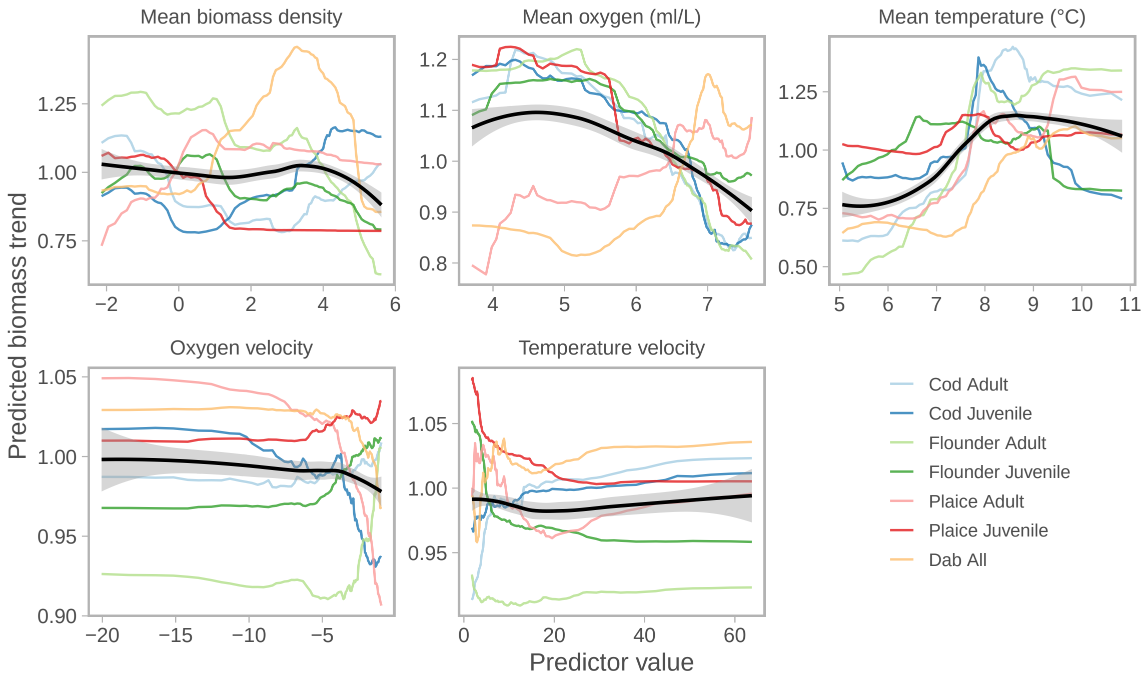

Figure S3: Partial dependence plots for all predictor variables (panels) in the random forest model of biomass trends. Colored lines represent individual species-age groups, and the black smoothed line shows the overall average trend across groups. Values are values are scaled relative to each group's mean prediction for easier comparison across groups.

References

Dunn, P. K., & Smyth, G. K. (1996). Randomized Quantile Residuals. Journal of Computational and Graphical Statistics, 5(3), 236–244. https://doi.org/10.2307/1390802

Hartig F (2022). DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models. R package version 0.4.6, https://CRAN.R-project.org/package=DHARMa.